

SensePi MPU

Installation and start-up guide for your laptop

SensePi is the app you use to measure how a model structure vibrates. It connects over Wi-Fi to a Raspberry Pi with **four motion sensors** attached, and lets you watch the signals live, record them, find the structure's natural frequencies and compare them with a computer model.

This guide takes you from the SensePi ZIP file to a running app, one step at a time. Type the commands exactly as shown. Where Windows and macOS differ, both are given: follow the one for your laptop.

Before you start, you need

- A laptop with **Windows 10 or 11**, or **macOS**.
- An **internet connection** for the installation (Steps 1 to 4).
- About **2 GB of free disk space** and about **20 minutes**, most of it waiting for downloads.
- The **SensePi ZIP file** from your instructor.
- For Step 6, later: your Raspberry Pi's **IP address** and **password**, from your instructor.

The steps

	Step	What you do
1	Install Python 3.12	Install the programming language SensePi is written in
2	Unzip SensePi	Extract the ZIP file into a folder
3	Open a terminal	Open a command window inside that folder
4	Install SensePi's packages	Let Python download everything SensePi needs
5	Start SensePi	Open the app
6	Connect to your Raspberry Pi	Enter the Pi's details in the app
7	Your first measurement	Watch and record the sensor signals

Steps 1 to 4 are done **once**. After that, starting SensePi takes two commands (see Starting SensePi again later).

STEP 1

Install Python 3.12

Use Python 3.12, not the newest version

The big yellow **Download Python** button on [python.org](https://www.python.org) gives you the newest Python, which SensePi has not been tested with. Use the link below to get **Python 3.12.10**. If your laptop already has a different Python, that's fine: install 3.12 next to it.

Windows

1. Open this page in your web browser: <https://www.python.org/downloads/release/python-31210/>
2. Scroll down to **Files** and click **Windows installer (64-bit)**.
3. Open the downloaded file. On the first screen, **tick both boxes at the bottom**: Use admin privileges when installing py.exe and Add python.exe to PATH. Then click **Install Now**.
4. When it finishes, click **Disable path length limit** if you see it (skip it if your laptop doesn't allow it), then click **Close**.
5. Check it worked: press the Start key, type **PowerShell**, open it, and type:

Windows (PowerShell)

```
py -3.12 --version
```

It should print `Python 3.12.10`. Then close PowerShell.

macOS

1. Open <https://www.python.org/downloads/release/python-31210/> and click **macOS 64-bit universal2 installer** under **Files**.
2. Open the downloaded file and click through the installer.
3. Check it worked: open **Terminal** (press Cmd+Space, type Terminal, press Return) and type:

macOS (Terminal)

```
python3.12 --version
```

It should print `Python 3.12.10`.

STEP 2

Unzip SensePi

Windows

1. Find the SensePi ZIP file, usually in your **Downloads** folder.
2. Right-click it and choose **Extract All....** Click **Browse...**, choose **Desktop**, then click **Select Folder** and **Extract**.
3. Open the new folder on your Desktop. If it contains just one more folder, open that too. Stop when you see `main.py` and `requirements.txt`: this is your **SensePi folder**.

macOS

1. Double-click the ZIP file in **Downloads**. A folder appears next to it.

2. Drag that folder to your **Desktop**. Open it until you see `main.py` and `requirements.txt`: this is your **SensePi folder**.

Always extract the ZIP first

Windows lets you look inside a ZIP file without extracting it, but SensePi can't run from inside a ZIP. Also keep the folder somewhere simple like the Desktop: a folder buried many levels deep can make Step 4 fail with "The filename or extension is too long".

Windows may hide file endings, so `main.py` can appear as just `main`.

STEP 3

Open a terminal in the SensePi folder

A terminal is a window where you type commands. It must be opened **inside the SensePi folder**, so the commands find SensePi's files.

Windows

- **Windows 11:** in File Explorer, open your SensePi folder, right-click an empty space and choose **Open in Terminal**.
- **Windows 10:** in File Explorer, open your SensePi folder, click the address bar at the top, type `powershell` and press Enter.

macOS

- Open **Terminal**, type `cd` followed by a space, drag your SensePi folder from Finder into the Terminal window, and press Return.

Check that you're in the right place:

Windows

```
dir main.py
```

macOS

```
ls main.py
```

You should see `main.py` listed. If you get "cannot find" or "No such file", the terminal is in the wrong folder: close it and repeat this step.

STEP 4

Install SensePi's packages

Type these commands one at a time, pressing Enter after each. Wait for each to finish (the line with the blinking cursor comes back) before typing the next.

Windows (PowerShell)

```
py -3.12 -m venv .venv
.venv\Scripts\Activate.ps1
python -m pip install --upgrade pip
pip install -r requirements.txt
```

macOS (Terminal)

```
python3.12 -m venv .venv
source .venv/bin/activate
python -m pip install --upgrade pip
pip install -r requirements.txt
```

What these do:

- The **first** command makes a private Python environment for SensePi: a folder called `.venv` inside your SensePi folder. Nothing else on your laptop is changed.
- The **second** switches it on. From now on, your command line starts with `(.venv)`.
- The **last** command downloads and installs everything SensePi needs (a few hundred MB). This takes several minutes. It is finished when you see `Successfully installed ...` and the command line comes back.

Windows: "running scripts is disabled on this system"?

If the second command shows this red message, type the following once, answer **Y**, then type `.venv\Scripts\Activate.ps1` again:

```
Set-ExecutionPolicy -Scope CurrentUser RemoteSigned
```

STEP 5

Start SensePi

In the same terminal (it still shows `(.venv)`), type:

Windows and macOS

```
python main.py
```

The SensePi window opens. The first start can take up to a minute.



The SensePi window. The tabs along the top are **Live Signals**, **Spectrum**, **Model Updating**, **Sonification**, **Digital Twin Experiment** and **Settings**.

Keep the terminal open

SensePi runs from the terminal. Closing the terminal closes SensePi. Messages that appear in the terminal while SensePi runs are normal.

STEP 6

Connect to your Raspberry Pi

Connect your laptop to the MissionControl Wi-Fi first

The Raspberry Pis can only be reached on the **MissionControl** Wi-Fi network. Laptops sometimes switch back to another saved network by themselves, so check this whenever SensePi can't connect.

Enter the Pi's details

1. Click the **Settings** tab.
2. Under **Configured Pis**, click **ExamplePi**.
3. Fill in the fields on the right as in the table below. Type the paths exactly as shown.
4. Click **Save host config**.

Field	What to type
Name	Any name you like, for example Pi-1
Host / IP	The Pi's IP address from your instructor (the Pi's small screen also shows it)
User	<code>verwalter</code>
SSH port	<code>22</code>
Password	The password from your instructor
Scripts base path	<code>/home/verwalter/sensor4</code>
Data directory	<code>/home/verwalter/logs</code>
Pi config path	<code>/home/verwalter/sensor4/pi_config.yaml</code>

The Pi details filled in. Replace “IP from your instructor” with the real IP address.

Check the sensor settings

1. Further down, under **Sensor defaults**, check that **Number of sensors** is **4 sensors** (it is by default).

Number of sensors: 4 sensors.

2. In the **Sensor placement map**, set each sensor's **Floor** and **Location** to where it is on your structure. Floor **0** is the base (the shaker). Every other tab uses this map, so ask your instructor if you're not sure.

Sensor	Floor	Location	Role
Sensor 1	0 - base / shaker	B2	base
Sensor 2	1	B2	structure
Sensor 3	2	B2	structure
Sensor 4	3	B2	structure

3 structural sensor(s) on floor(s) 1, 2, 3 - base sensor present (measures input)
Used by every tab: Smart Recording, Spectrum, Model Updating, Sonification and the Digital Twin all read this map.

The sensor placement map, set here for a base sensor plus one sensor on each of three floors.

3. Click **Save sensors.yaml**.
4. Click **Sync Pi defaults (pi_config.yaml)** to send these settings to the Pi.

STEP 7

Your first measurement

Recording / stream controls

Sensor: MPU6050

Refresh: 50 Hz ☒ Normalise (centre on zero) Session:

Start

Stop

Rec length: 60 s Rate: 100 Hz

Smart Record

Sync logs from Pi

The recording controls at the top of the Live Signals tab.

1. Click the **Live Signals** tab, then **Start**.
2. Tap the structure gently and watch the signals move. There is one row of plots per sensor.
3. Click **Stop** when you're done.

To keep a measurement

1. Set **Rec length** to how many seconds you want to record. Record **at least 30 seconds** if you want to analyse it later in **Model Updating**.
2. Optionally type a **Session** name, for example team3_test1.
3. Click **Smart Record**. SensePi first checks the real sampling rate for a few seconds, then records for the length you chose and saves the file by itself.

Starting SensePi again later

You only do Steps 1 to 4 once. Next time, open a terminal in your SensePi folder (Step 3) and type:

Windows (PowerShell)

```
.venv\Scripts\Activate.ps1
python main.py
```

macOS (Terminal)

```
source .venv/bin/activate
python main.py
```

Your Pi details and sensor map are remembered, so you don't have to enter them again.

Where your data is saved

Everything SensePi saves goes into the `output` folder inside your SensePi folder:

Folder	What's inside
output/sensor_recordings	Your recorded measurements, one folder per recording
output/model	Computer-model files from Model Updating
output/digital_twin	Digital Twin experiment runs
output/sonification	Sound you captured

At the end of the camp

Copy the `output` folder somewhere safe **before** you delete the SensePi folder. To remove SensePi completely, just delete the SensePi folder.

Troubleshooting

Problem	What to do
<code>py</code> or <code>python3.12</code> is "not recognized" / "command not found", or typing <code>python</code> opens the Microsoft Store	Python 3.12 isn't installed, or Add <code>python.exe</code> to PATH wasn't ticked. Run the installer from Step 1 again, tick both boxes, then open a new terminal.
"running scripts is disabled on this system" (Windows)	Run <code>Set-ExecutionPolicy -Scope CurrentUser RemoteSigned</code> , answer Y , then run the activate command again.
Step 4 fails with "The filename or extension is too long"	Your SensePi folder is too deep. Move it to your Desktop, delete the <code>.venv</code> folder inside it, and do Step 4 again.
<code>pip install</code> says "No matching distribution found"	The environment was made with the wrong Python. Delete the <code>.venv</code> folder, check Step 1 (<code>py -3.12 --version</code>), and do Step 4 again.
SensePi says "matplotlib is not installed" or "No module named ..."	The environment isn't switched on: your command line must start with <code>(.venv)</code> . Run the activate command, then <code>python main.py</code> again. If it still fails, run <code>pip install -r requirements.txt</code> again.
SensePi can't connect to the Pi	Check that the laptop is on the MissionControl Wi-Fi, and check the IP address and password in Settings . Ask your instructor whether the Pi is switched on (its screen shows its IP address).
No signal, or a sensor is missing	In Settings , check Number of sensors is 4 sensors , then click Sync Pi defaults (pi_config.yaml) again. Check the sensor cables, then tell your instructor.
Plots are slow or jerky	Close other programs you don't need, or set Refresh lower in Live Signals.

Quick reference

Keep this page handy. All commands are typed in a terminal opened in your SensePi folder.

	Windows (PowerShell)	macOS (Terminal)
Check Python	<code>py -3.12 --version</code>	<code>python3.12 --version</code>
Make the environment (once)	<code>py -3.12 -m venv .venv</code>	<code>python3.12 -m venv .venv</code>
Switch it on	<code>.venv\Scripts\Activate.ps1</code>	<code>source .venv/bin/activate</code>
Install packages (once)	<code>pip install -r requirements.txt</code>	<code>pip install -r requirements.txt</code>
Start SensePi	<code>python main.py</code>	<code>python main.py</code>

Setting on the Pi	Value
Wi-Fi network	MissionControl
User	<code>verwalter</code>
SSH port	<code>22</code>
Scripts base path	<code>/home/verwalter/sensor4</code>
Data directory	<code>/home/verwalter/logs</code>
Pi config path	<code>/home/verwalter/sensor4/pi_config.yaml</code>
IP address and password	From your instructor

The tabs

Tab	What it does
Live Signals	Streams the four sensors live and records measurements
Spectrum	Shows the frequencies the structure vibrates at (natural frequencies and mode shapes)
Model Updating	Tunes a computer model of the structure to match your measurements
Sonification	Turns the vibrations into sound
Digital Twin Experiment	Runs the computer model alongside the real structure and compares them
Settings	Raspberry Pi connection, number of sensors, sensor positions and sampling rate